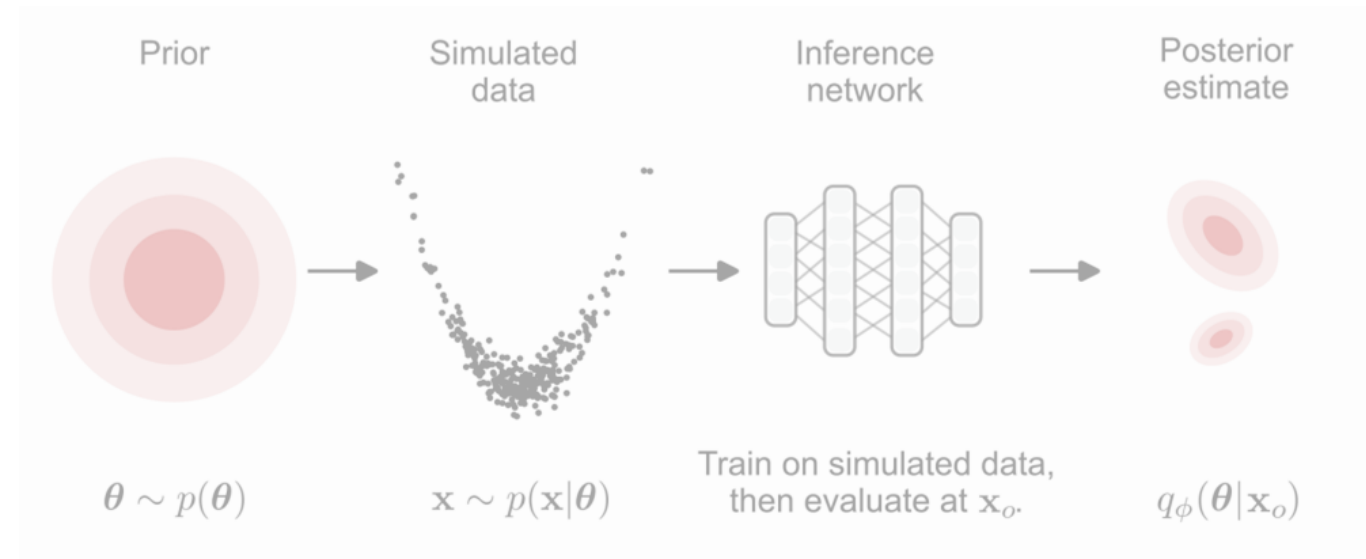


# SIMULATION BASED INFERENCE

Presentation by Jakob Dammann

Scientific Machine Learning Seminar 2026



# Forward vs Inverse Problems

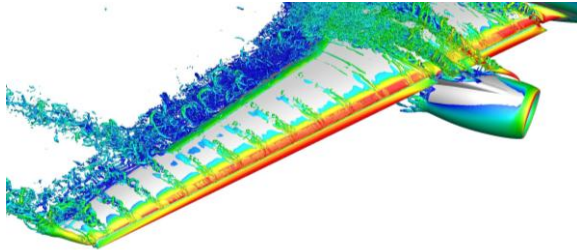


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Previously (**Forward**)

solve ODEs, simulations

$x \sim p(x|\theta)$  or  $p(x|\theta)$

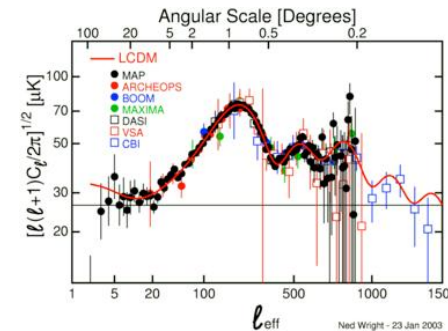


data:  $x$     parameters:  $\theta$

Now (**Inverse**)

parameters from observations

$p(\theta|x)$

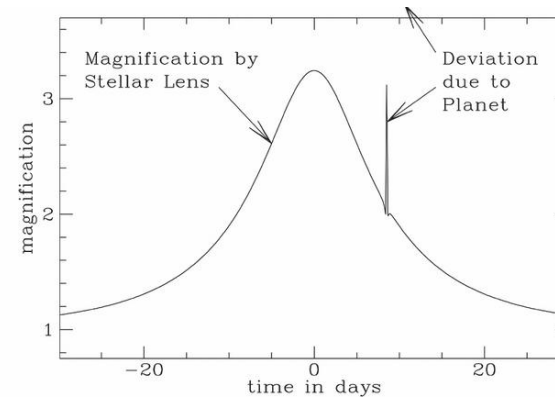
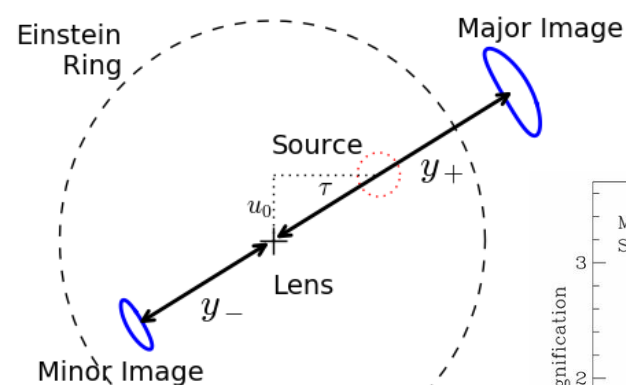
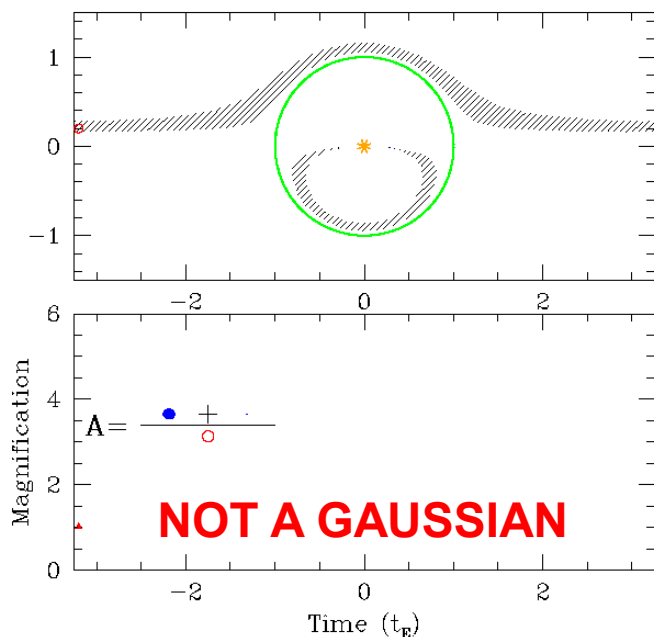
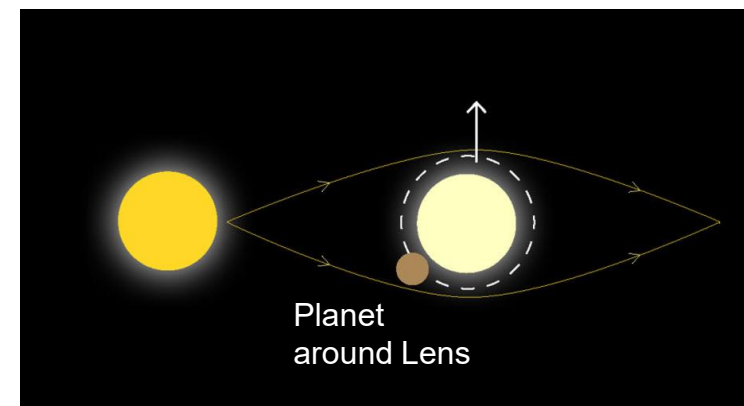
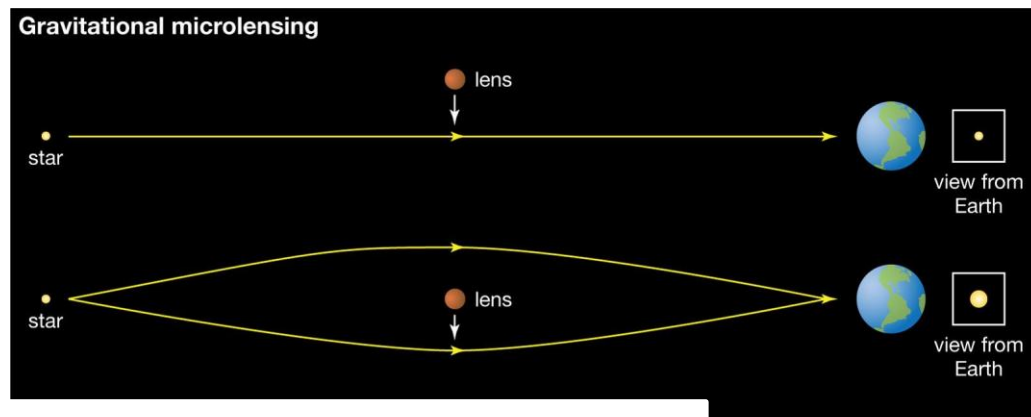


$$p(\theta|x) = \frac{p(x|\theta)}{p(x)} p(\theta)$$

# Gravitational Microlensing as an Inverse Problem



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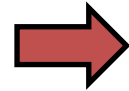
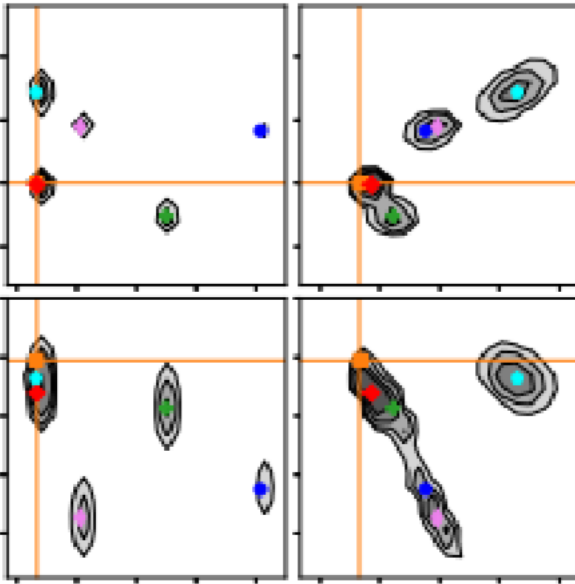


# Struggles of Classical Bayesian Inference



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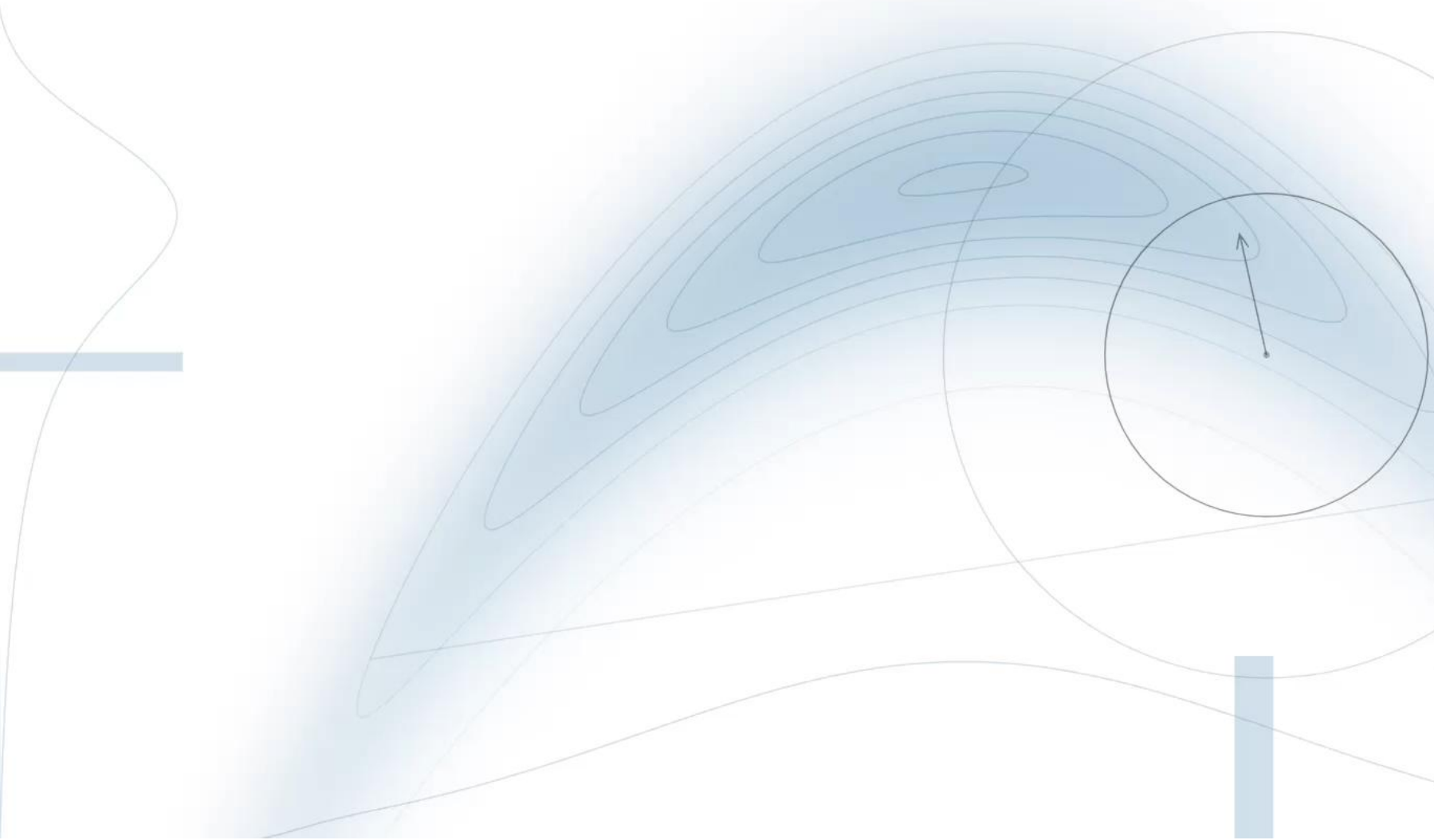
**Multimodal** parameter distribution:



use **MCMC** to converge to a posterior

$$p(\theta|x) = \frac{p(x|\theta)}{p(x)} p(\theta)$$

Random walk Metropolis-Hastings



Simulation options

Algorithm

RandomWalkMH

▼

Target distribution

banana

▼

Autoplay

☒

Autoplay delay

250

Tweening delay

0

Step

Reset

Visualization Options

Animate proposal

☒

Show target

☒

Show samples

☒

Show histogram

☒

Histogram bins

50

Algorithm Options

Proposal  $\sigma$

1

About this algorithm

Close Controls

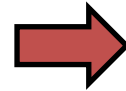
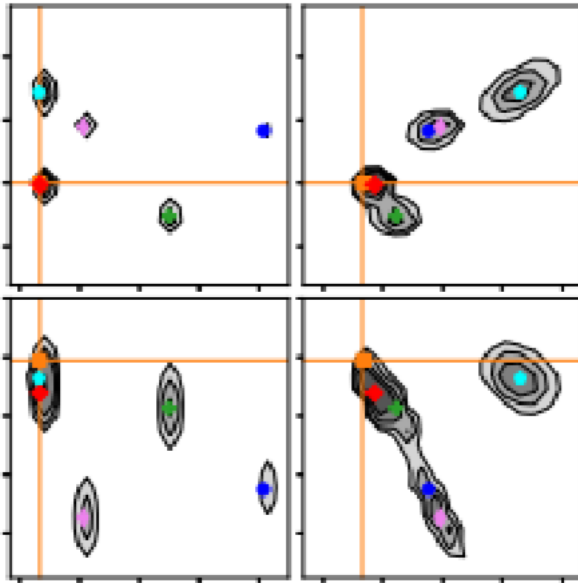
$$p(\theta|x) = \frac{p(x|\theta)}{p(x)} p(\theta)$$

# Struggles of Classical Bayesian Inference



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**Multimodal** parameter distribution:



use **MCMC** to converge to a posterior

but:

- convergence computationally **expensive**
- likelihood expensive or **approximated** locally with Gaussian
- **bad at modelling** multimodal posteriors, degeneracies

$$p(\theta|x) = \frac{p(x|\theta)}{p(x)} p(\theta)$$

# Simulation Based Inference



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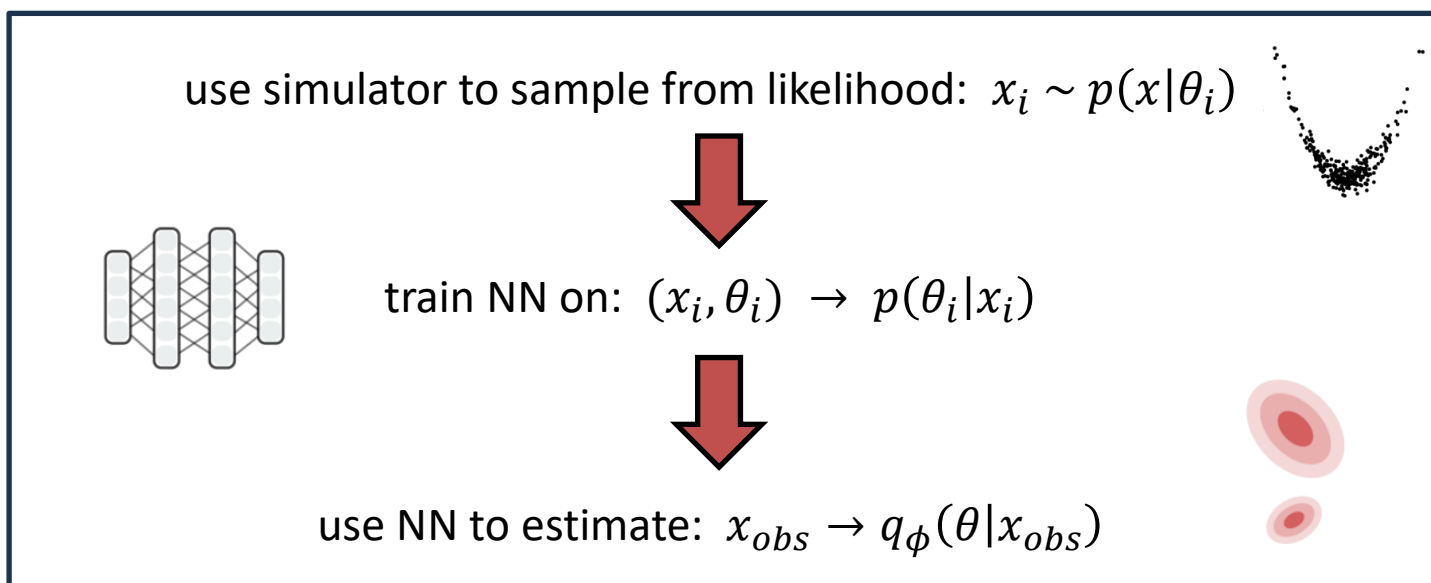
Key Idea:

**skip likelihood evaluation** --> directly relate  $x$  to  $p(\theta|x)$

likelihood **implicit** in a simulator

$$p(\theta|x) = \frac{p(x|\theta)}{p(x)} p(\theta)$$

## Neural Posterior Estimation (NPE)

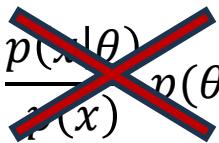


Variations:

**Neural Likelihood  
Estimation (NLE)**

**Neural Likelihood Ratio  
Estimation (NRE)**

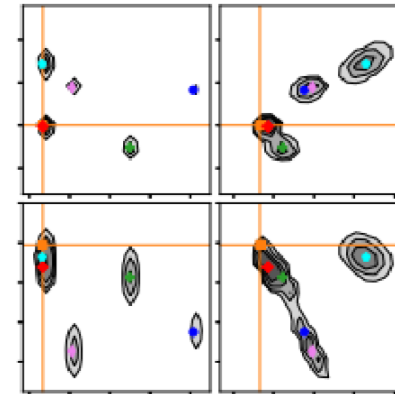
## Likelihood-free

$$p(\theta|x) = \frac{p(x|\theta)}{p(x)} p(\theta)$$


## Amortized Inference

„Pay computation cost  
upfront.“

## Multimodal Posteriors

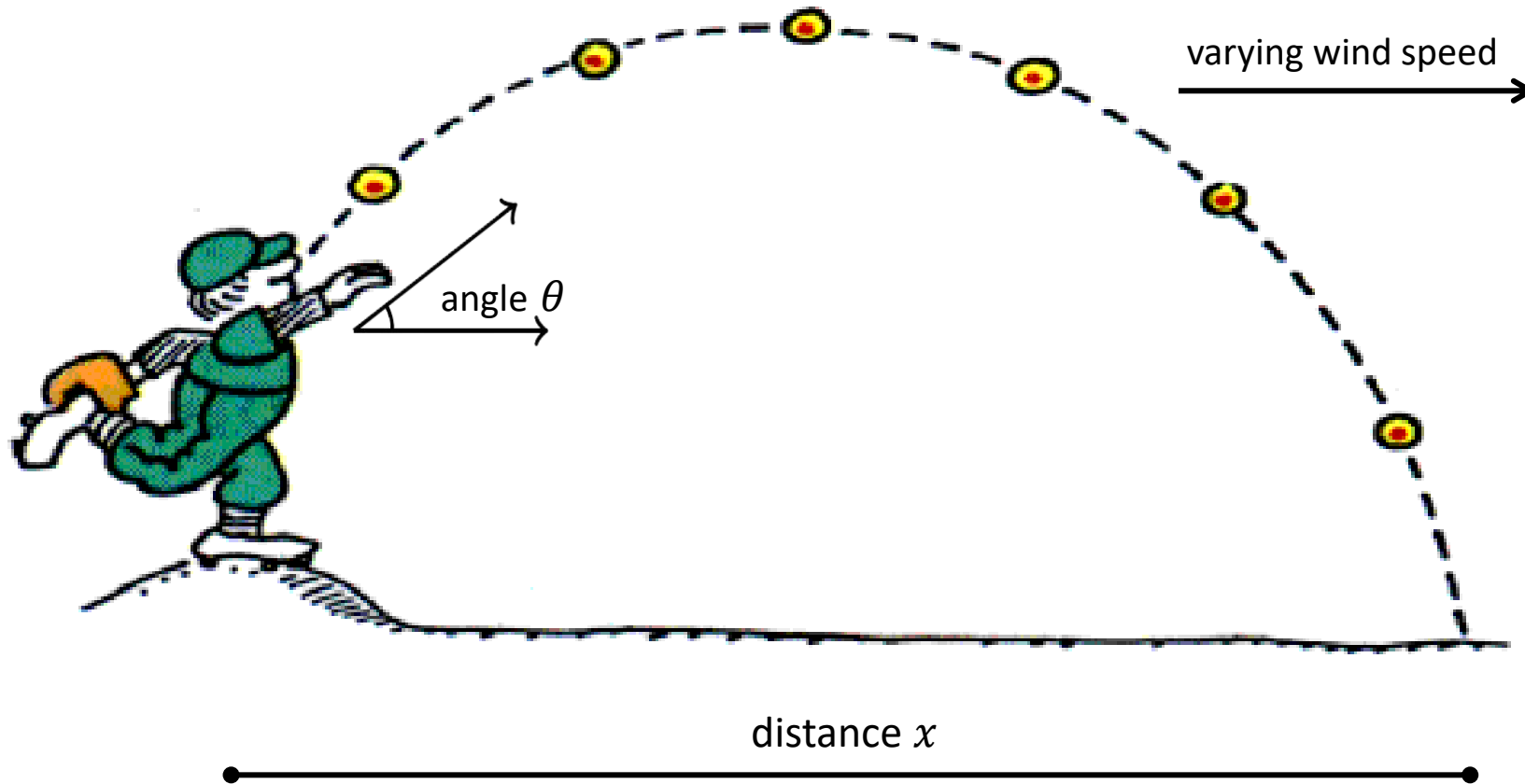


# Toy Example

adapted from Deistler et al. (2025)



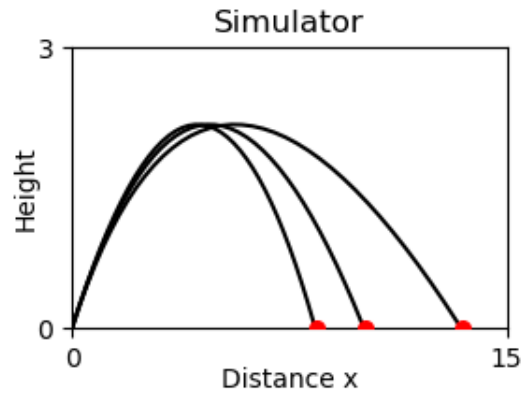
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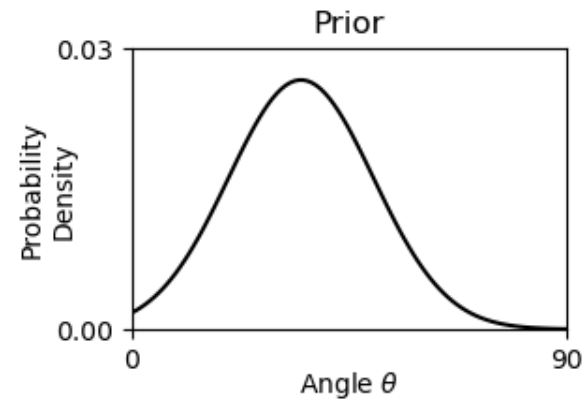
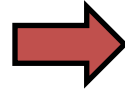
# Using SBI: Simulation



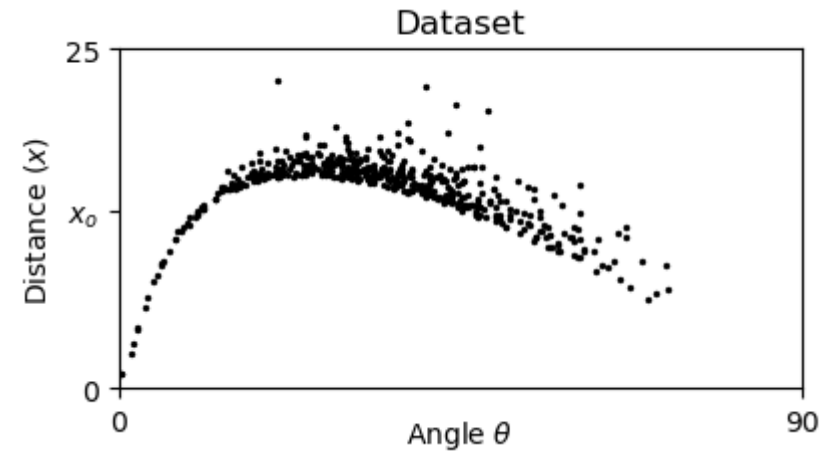
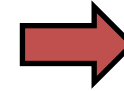
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create a simulator that  
can sample:  $x_i \sim p(x|\theta_i)$



define prior:  $p(\theta)$

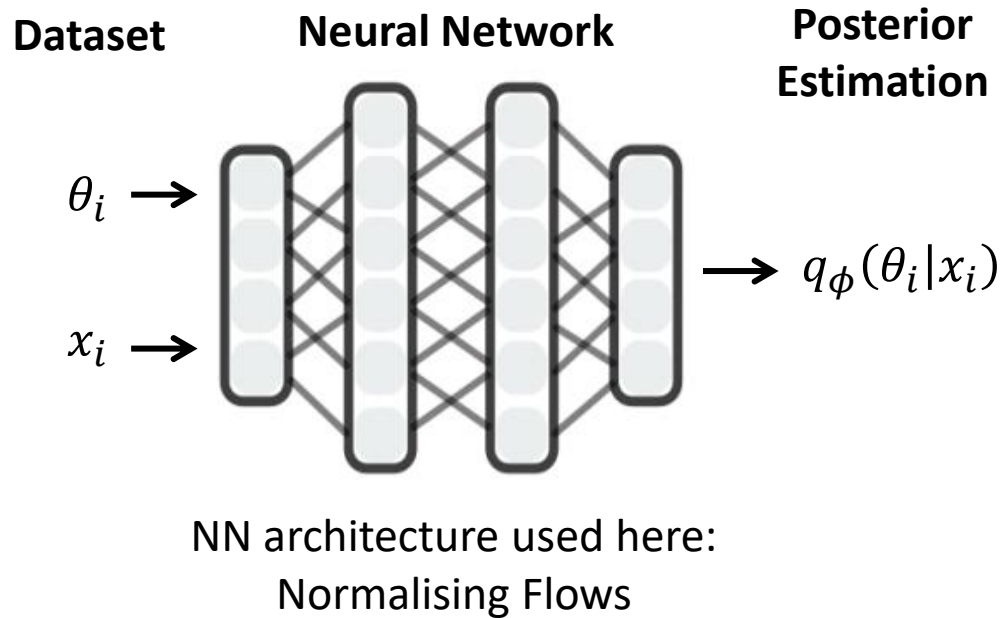


sample from prior  $\theta_i \sim p(\theta)$   
run simulator to  $x_i \sim p(x|\theta_i)$   
to create dataset  $(\theta_i, x_i)$

# Using SBI: Training



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**Loss Function:**

$$\mathcal{L}(\phi) = \mathbb{E}_{(\theta_i, x_i) \sim p(\theta, x)} [-\log q_\phi(\theta_i | x_i)]$$

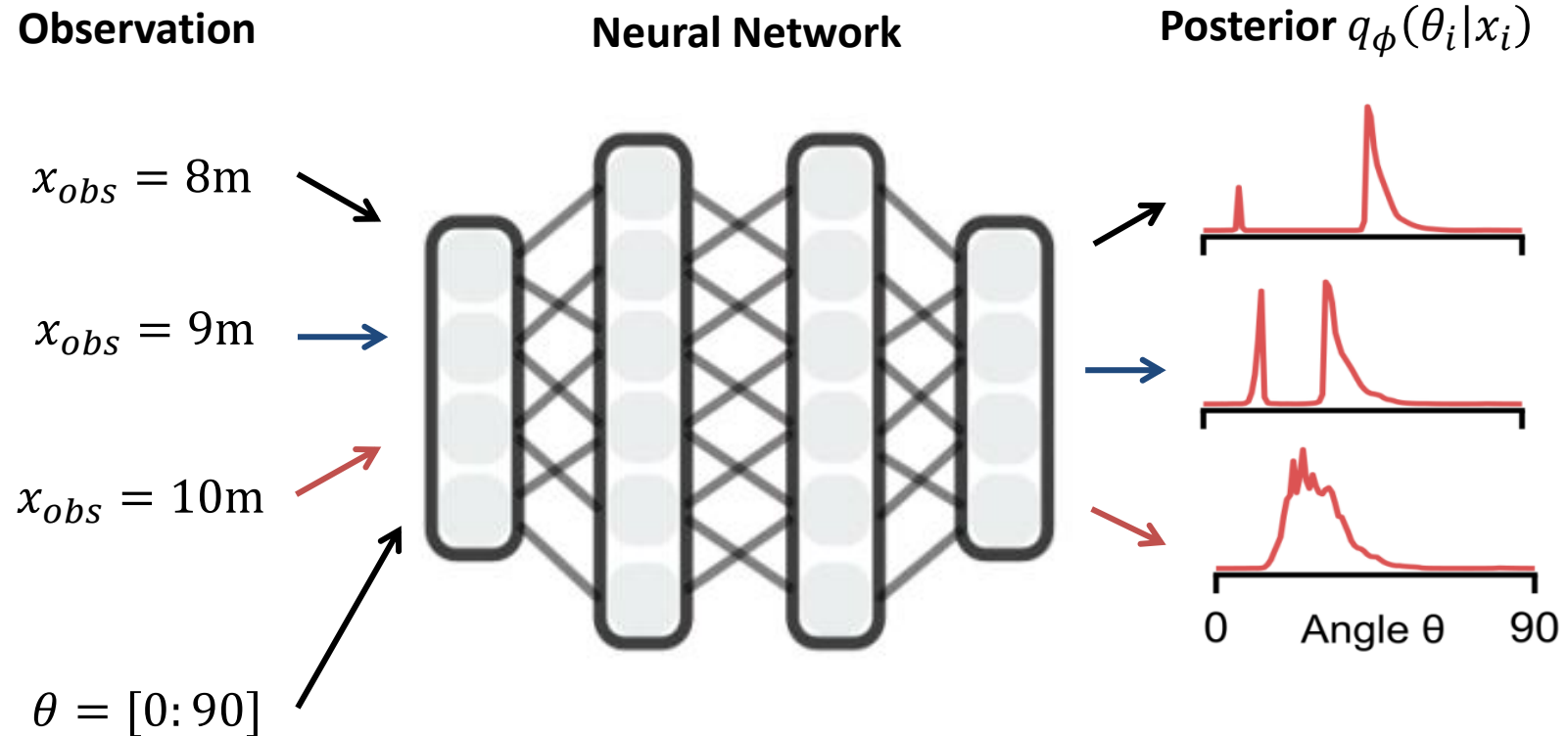
cross entropy:

$$-\int p(\theta | x_i) \log q_\phi(\theta | x_i) d\theta = -\log q_\phi(\theta_i | x_i)$$

# Using SBI: Evaluation



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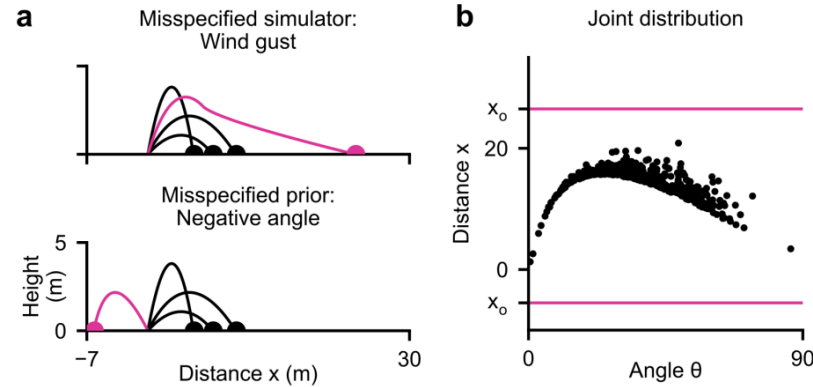
# Code Example

# Limitations of SBI



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- misspecification: NN gives nonsensical posterior



- posterior is **neural approximation**, not analytical/numerical
- quality check and **calibration of posterior** difficult

Benefits:

Likelihood-free

Amortized Inference

Multimodal Posteriors

THE ASTRONOMICAL JOURNAL, 161:262 (11pp), 2021 June

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<https://doi.org/10.3847/1538-3881/abf42e>



## Real-time Likelihood-free Inference of Roman Binary Microlensing Events with Amortized Neural Posterior Estimation

Keming Zhang (张可名)<sup>1</sup>, Joshua S. Bloom<sup>1,2</sup>, B. Scott Gaudi<sup>3</sup>, François Lanusse<sup>4</sup>, Casey Lam<sup>1</sup>, and Jessica R. Lu<sup>1</sup>

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<sup>3</sup>Department of Astronomy, The Ohio State University, Columbus, OH 43210, USA

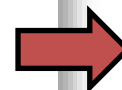
<sup>4</sup>AIM, CEA, CNRS, Université Paris-Saclay, Université Paris Diderot, Sorbonne Paris Cité, F-91191, Gif-sur-Yvette, France

Received 2021 February 10; revised 2021 March 22; accepted 2021 March 30; published 2021 May 11

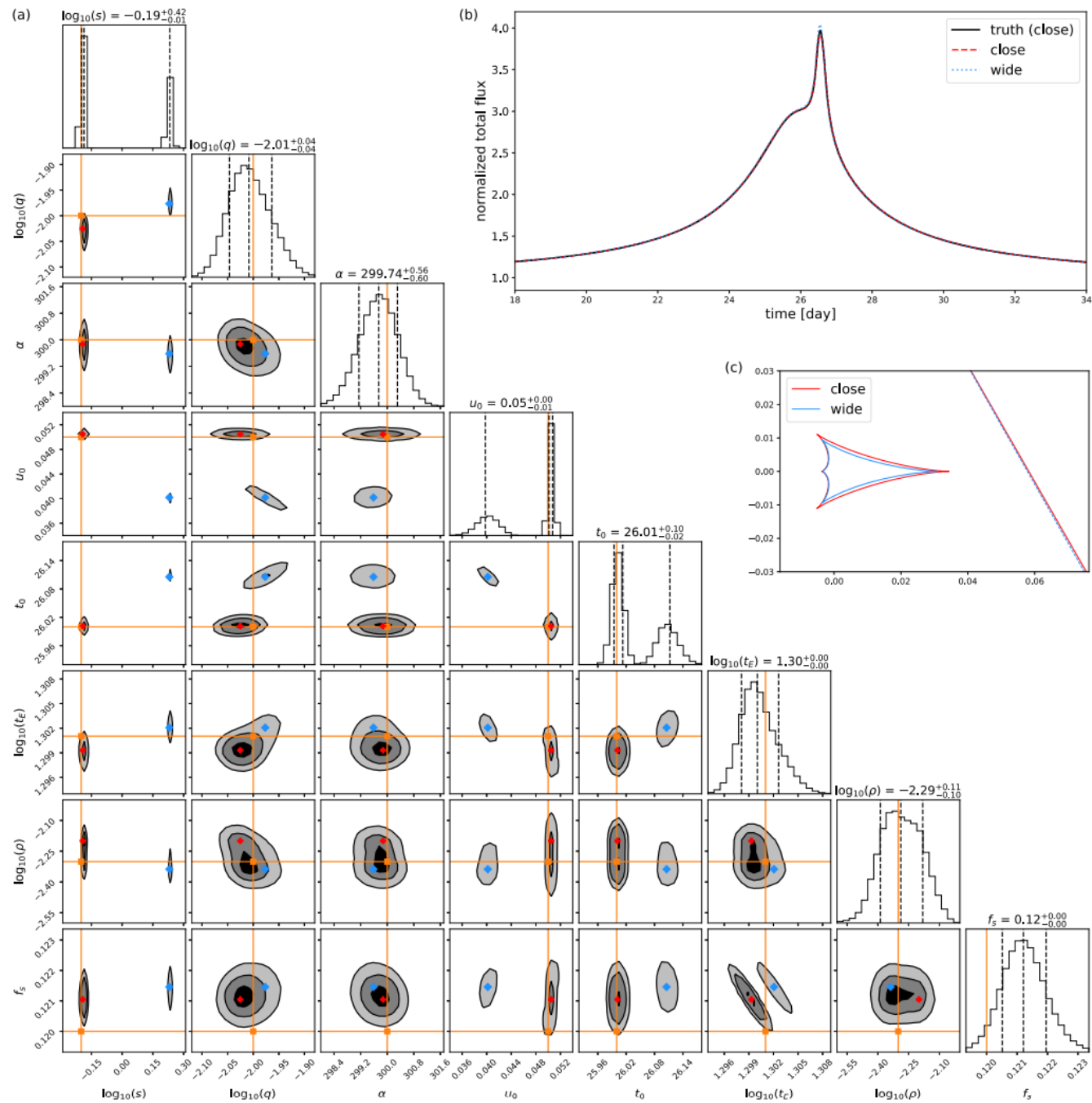
### Abstract

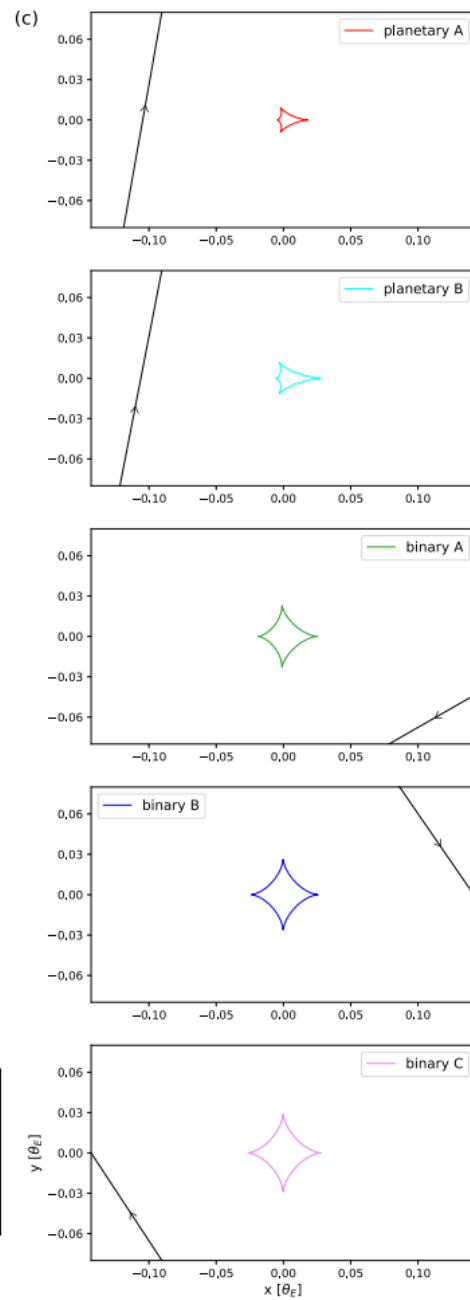
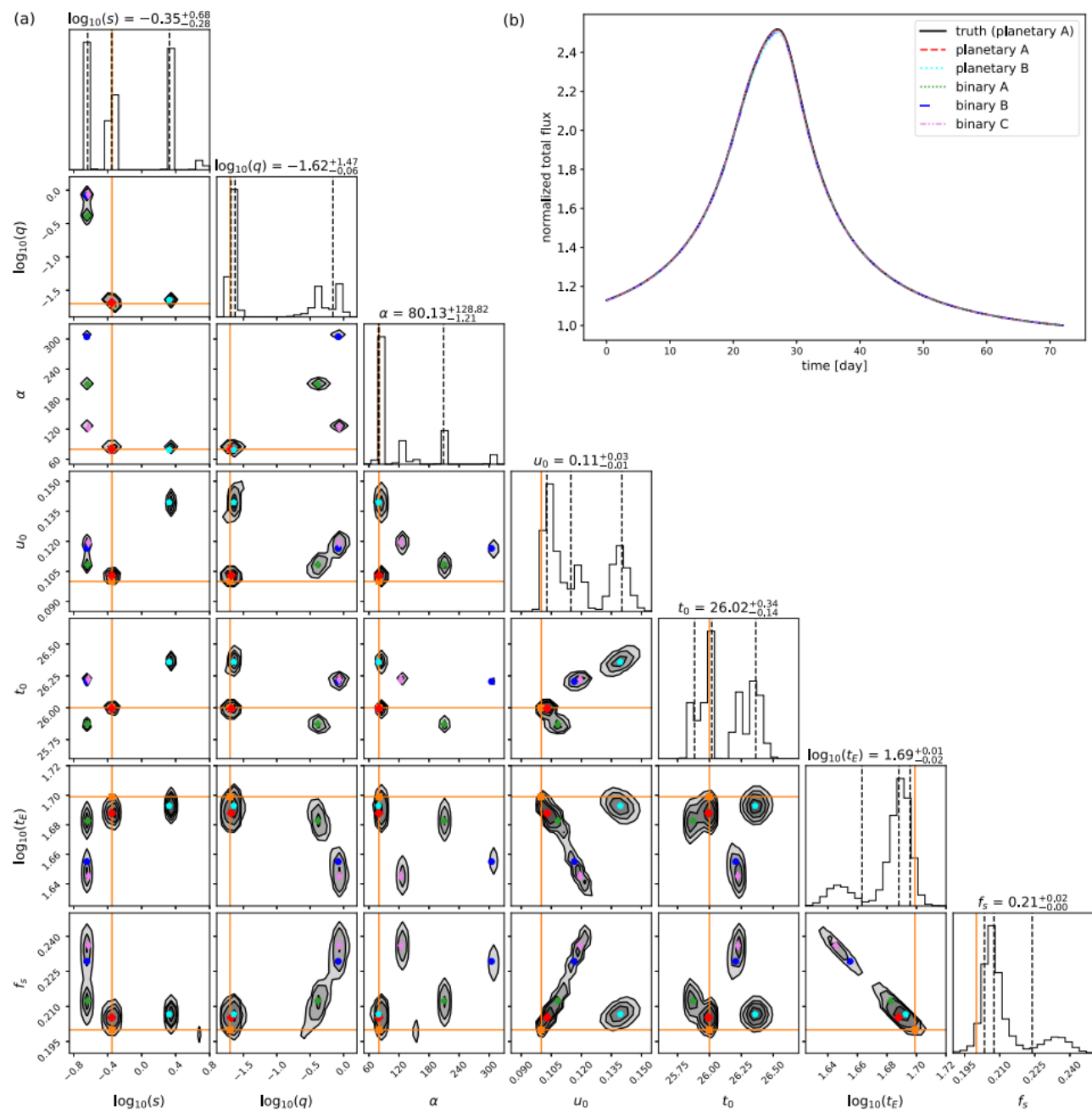
Fast and automated inference of binary-lens, single-source (2L1S) microlensing events with sampling-based Bayesian algorithms (e.g., Markov Chain Monte Carlo, MCMC) is challenged on two fronts: the high computational cost of likelihood evaluations with microlensing simulation codes, and a pathological parameter space where the negative-log-likelihood surface can contain a multitude of local minima that are narrow and deep. Analysis of 2L1S events usually involves grid searches over some parameters to locate approximate solutions as a prerequisite to posterior sampling, an expensive process that often requires human-in-the-loop domain expertise. As the next-generation, space-based microlensing survey with the Roman Space Telescope is expected to yield thousands of binary microlensing events, a new fast and automated method is desirable. Here, we present a likelihood-free inference approach named amortized neural posterior estimation, where a neural density estimator (NDE) learns a surrogate posterior  $\hat{p}(\theta|\mathbf{x})$  as an observation-parameterized conditional probability distribution, from pre-computed simulations over the full prior space. Trained on 291,012 simulated Roman-like 2L1S simulations, the NDE produces accurate and precise posteriors within seconds for any observation within the prior support without requiring a domain expert in the loop, thus allowing for real-time and automated inference. We show that the NDE also captures expected posterior degeneracies. The NDE posterior could then be refined into the exact posterior with a downstream MCMC sampler with minimal burn-in steps.

*Unified Astronomy Thesaurus concepts:* Binary lens microlensing (2136); Gravitational microlensing exoplanet detection (2147)



- trained NN for 25 hours on 300K simulations
- speed improvement:  
1 simulation / s  
↓  
10<sup>5</sup> posterior samples / s
- using **MCMC** after SBI to achieve precise solution



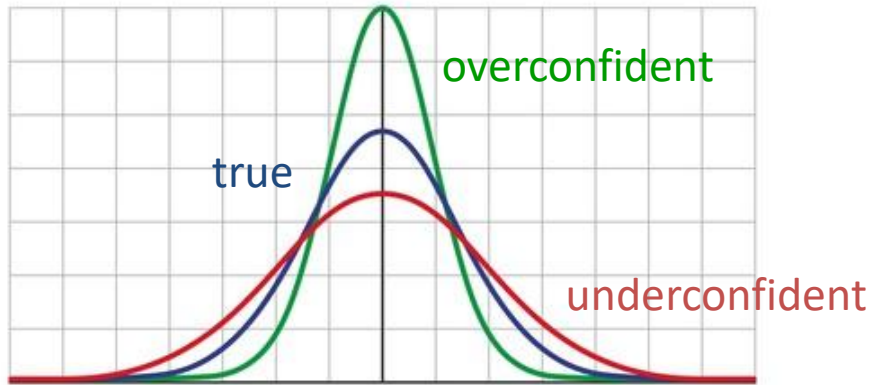


# SBI in Microlensing: Posterior Calibration

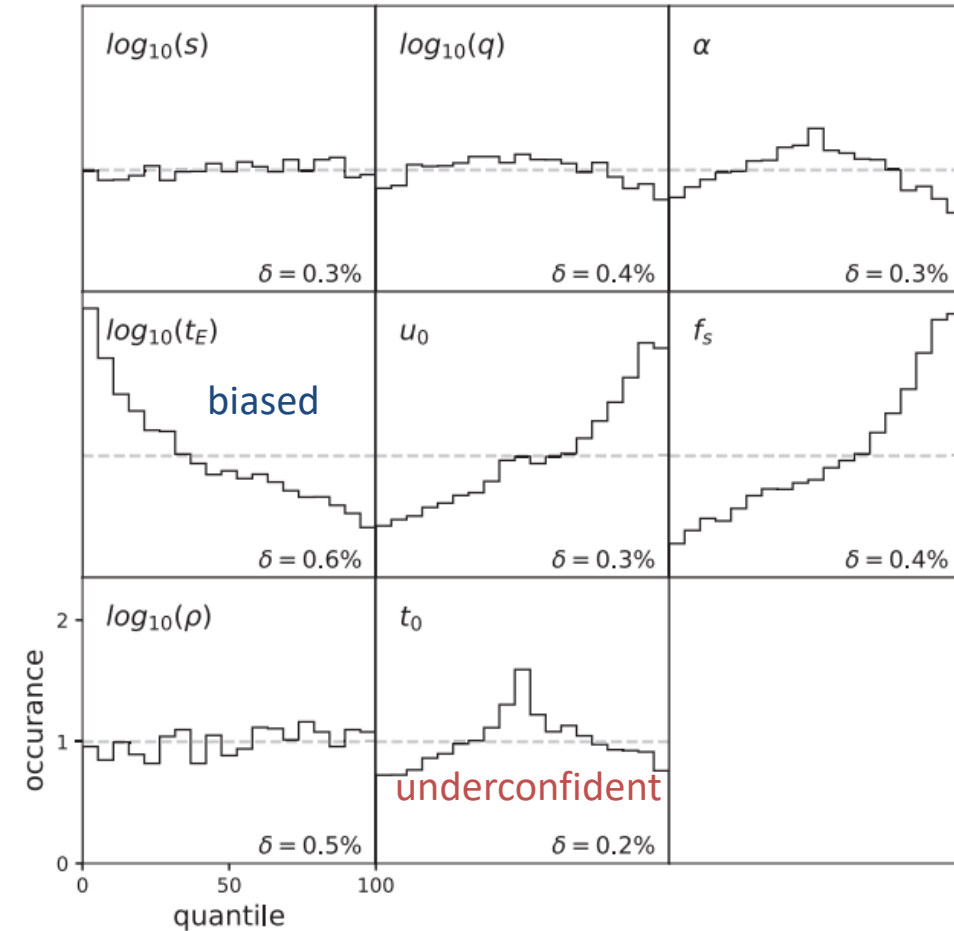


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example posterior:



Posterior Quantile Calibration:



## Flow Matching for Scalable Simulation-Based Inference

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Machine Learning in Science,  
University of Tübingen  
Tübingen, Germany

**Bernhard Schölkopf**  
Max Planck Institute for Intelligent Systems  
Tübingen, Germany

### Abstract

Neural posterior estimation methods based on discrete normalizing flows have become established tools for simulation-based inference (SBI), but scaling them to high-dimensional problems can be challenging. Building on recent advances in generative modeling, we here present flow matching posterior estimation (FMPE), a technique for SBI using continuous normalizing flows. Like diffusion models, and in contrast to discrete flows, flow matching allows for unconstrained architectures, providing enhanced flexibility for complex data modalities. Flow matching, therefore, enables exact density evaluation, fast training, and seamless scalability to large architectures—making it ideal for SBI. We show that FMPE achieves competitive performance on an established SBI benchmark, and then demonstrate its improved scalability on a challenging scientific problem: for gravitational-wave inference, FMPE outperforms methods based on comparable discrete flows, reducing training time by 30% with substantially improved accuracy. Our work underscores the potential of FMPE to enhance performance in challenging inference scenarios, thereby paving the way for more advanced applications to scientific problems.

replacing the Normalizing Flow NN with **Flow Matching NN (FMPE)**

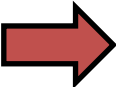
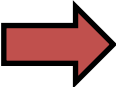
→ learns distribution as transformation of a **vector field**

- **fewer parameters** necessary, allows more complex architecture
- 30% training time reduction
- more expressive
- **slower inference** due to vector field

# Summary



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| Benefits                                | Limitations                                   |                                                                                                           |
|-----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>very fast</b> observation processing | upfront <b>Simulation</b> and <b>Training</b> |  Amortized Inference   |
| Likelihood-free                         | just Neural Approximation                     |                                                                                                           |
|                                         | very depended on simulation                   |                                                                                                           |
| able to model multimodal Posteriors     | Posterior may be uncalibrated                 |  Posterior Calibration |

So, when should I use SBI?

- inverse problem
- time critical
- expensive likelihood evaluation, multimodal posterior



- Michael Deistler, Jan Boelts, et al. Simulation Based Inference: A Practical Guide, 2025
- Keming Zhang, et al. Real-time Likelihood-free Inference of Roman Binary Microlensing Events with Amortized Neural Posterior Estimation. *The Astrophysical Journal*, 2021
- Maximilian Dax, Jonas Wildberger, Simon Buchholz, et al. Flow Matching for Scalable Simulation-Based Inference. 2023

# Appendix: Diagram



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